Worksheet & Answer Key

When white light passes through a glass prism, it is refracted, or bent, due to differences in speed between its component colours. This phenomenon is known as dispersion, where shorter wavelengths like violet and blue are bent more than longer wavelengths like red and orange. The dispersed colours then spread out and form a band of colour, which we perceive as the visible spectrum. The glass prism acts as a splitter, separating the different wavelengths of white light into their individual colours.

Word Bank:

prism, refracted, component, dispersion, wavelengths, visible, splitter, indivisible

Fill in the blanks:

1. When white light passes through a _____, it is _____ due to differences in speed between its _____ .

2. This phenomenon is known as _____, where shorter _____ like violet and blue are _____ more than longer _____ like red and orange.

3. The dispersed colours then spread out and form a band of ____ _____.

4. The glass ____ ____ acts as a _____, separating the different _____ of white light into their individual _____.

5. _____ wavelengths like violet and blue are bent more than longer ____ wavelengths like red and orange.

6. The visible _____ is what we perceive as the spread-out colours.

7. Differences in speed between its component _____ cause the light to be _____.

8. The glass prism separates the different _____ of white light into their individual _____.

■ Answer Key (for teachers only):

1. prism
2. refracted
3. component
4. dispersion
5. wavelengths
6. visible
7. splitter
8. indivisible